

Function-Oriented and Object-Oriented Software Design

Solved and Practice Exercises

UML Example – Veterinary System

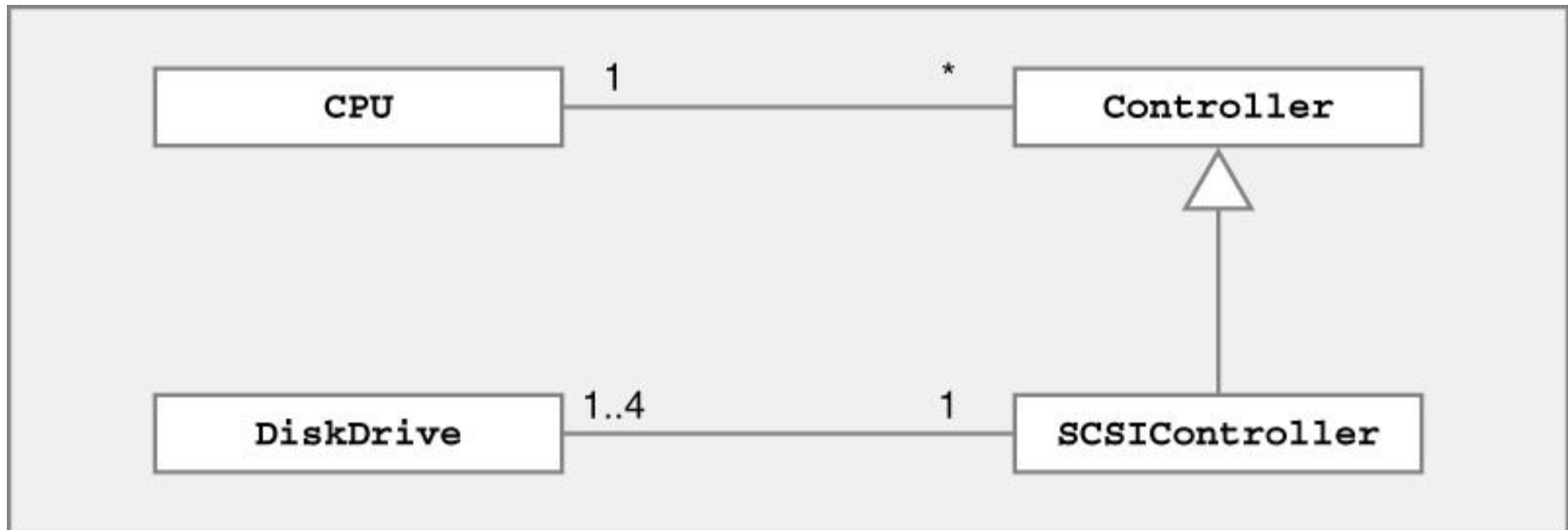
- Try to read & understand UML diagram



- 1 or more **Pets** associated with 1 **PetOwner**

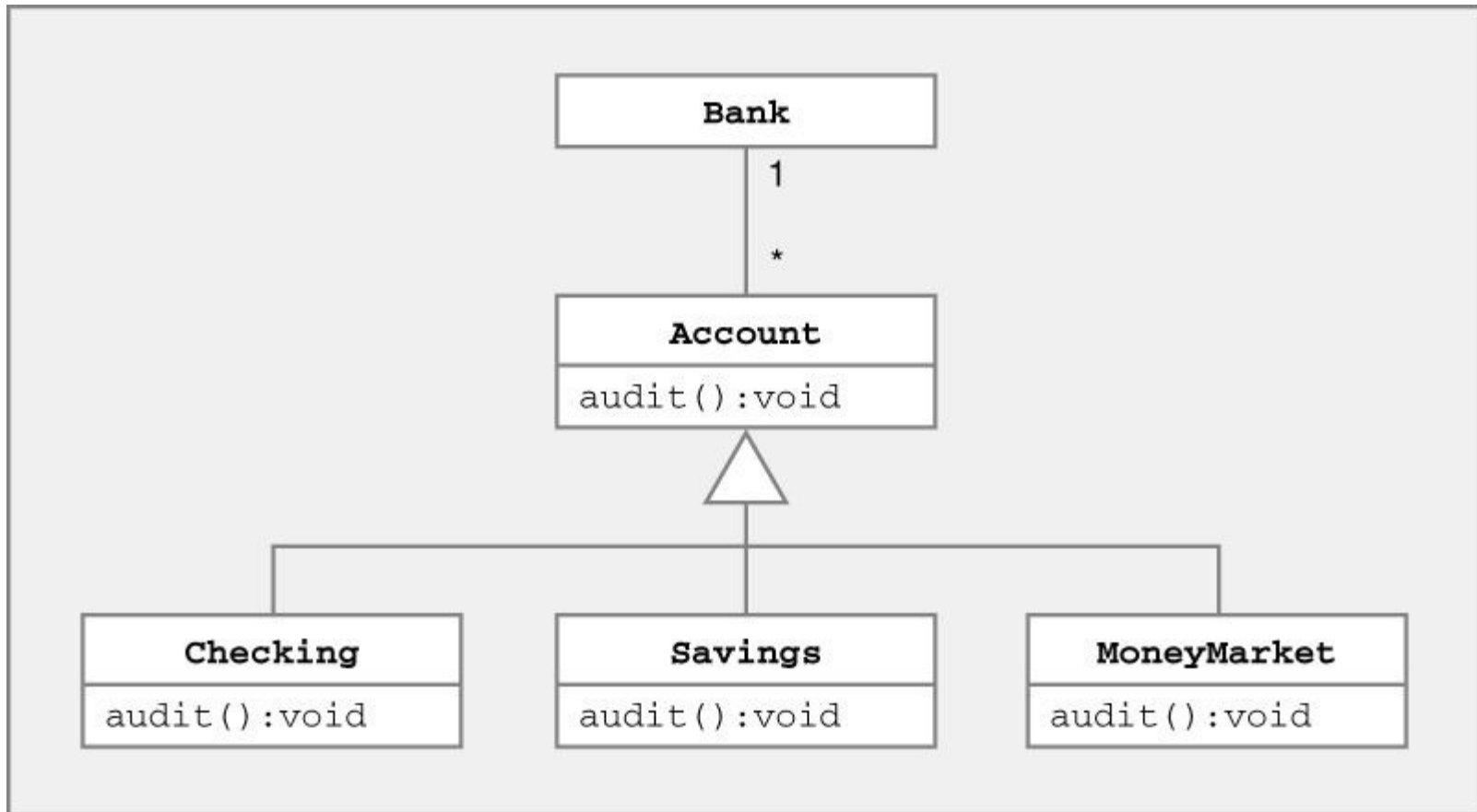
UML Example – Computer System

- Try to read & understand UML diagram



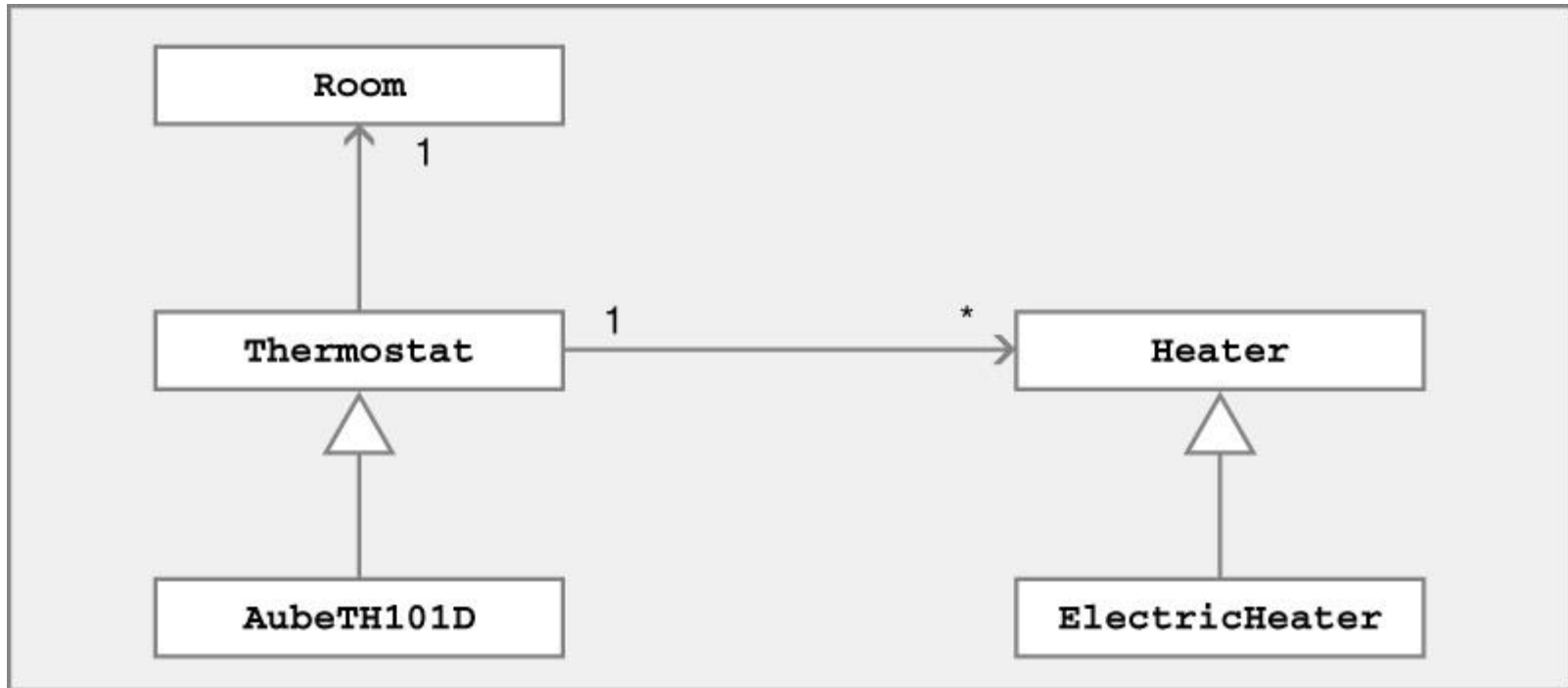
- 1 CPU associated with 0 or more Controllers
- 1-4 DiskDrives associated with 1 SCSIController
- SCSIController is a (specialized) Controller

UML Example – Banking System



- **1 Bank associated with 0 or more Accounts**
- **Checking, Savings, MoneyMarket are Accounts**

UML Example – Home Heating System



- Each Thermostat has 1 Room
- Each Thermostat associated with 0 or more Heaters
- ElectricHeater is a specialized Heater
- AubeTH101D is a specialized Thermostat

Exercise



Draw a use case diagram and a related scenario for the following situation:

- A user can borrow a book from a library;
 - extend it with borrowing a journal
- a user can give back a book to the library
 - including the use case when the user is identified

Exercise: include or extend?



Main use cases: a customer buys something (eg. a book) from a virtual store like Amazon

- The user must be identified
- The book is not currently available, delayed delivery
- When the book is received the service must be graded
- The book is delivered via air mail
- The book is an ebook and can be delivered via Internet

Do-it-Yourself Exercise

- Draw a use case diagram and a class diagram to represent the following system.

Over the summer holiday, university students can book college hall accommodation online. They must specify their name, student number, course, year, and identify three college residences as their preferences. The system makes an allocation of students to rooms before the start of the term, trying, where possible, to allocate students to a room in one of their preferred halls.

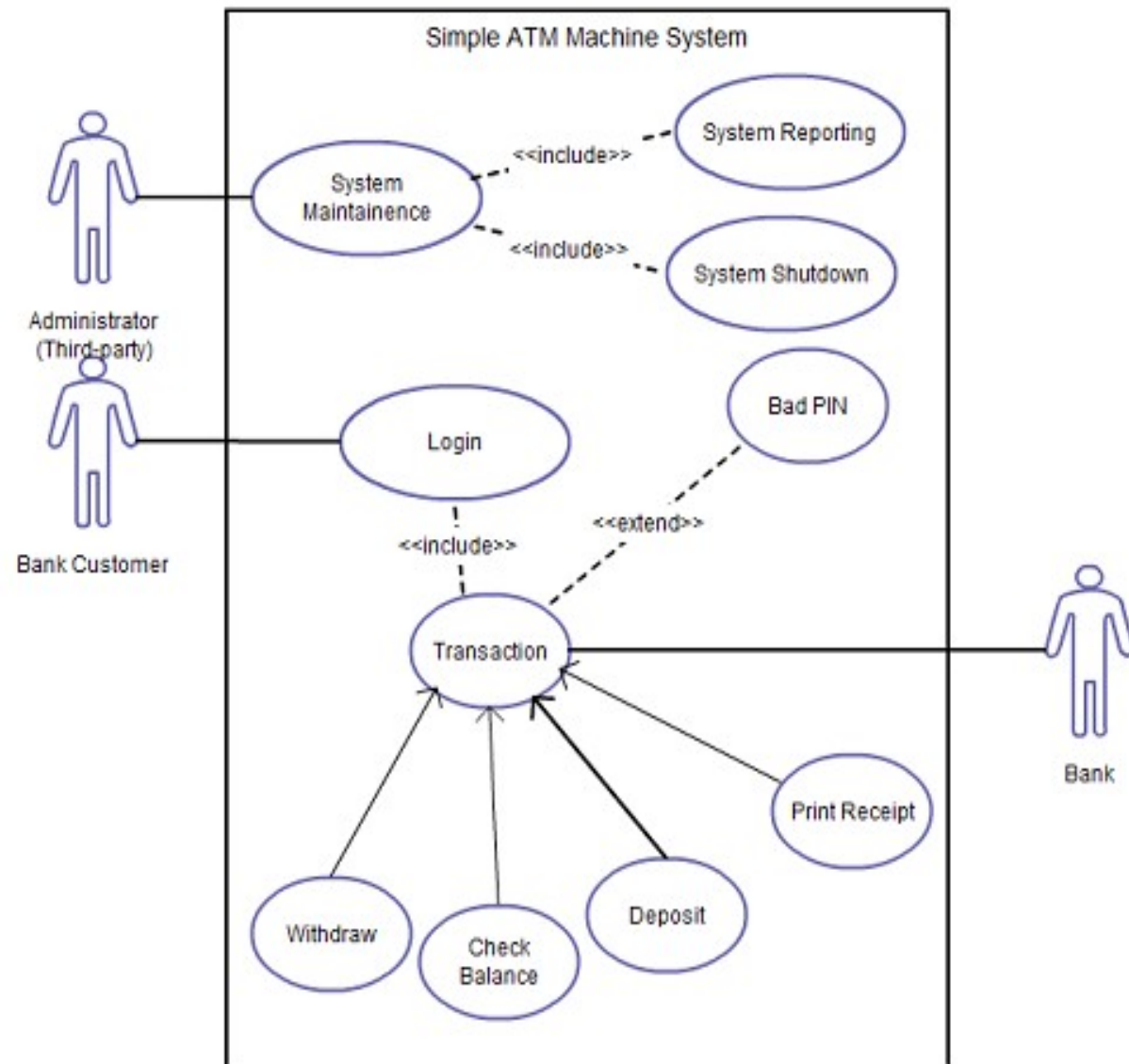
Exercise



Draw a sequence diagram showing how a customer interacts with a travel agency, a station and a train to reach some destination

Draw a sequence diagram to show how a user prints a document on a printer, and a counter keeps a count of printed pages

Design Use Case Model for an ATM Machine



Design Use Case Model

Let's imagine we will develop a browser-based training system to help people prepare for a certification exam. A user can request a quiz to the system. The system picks a set of questions from its database, and compose them together to make a quiz. It rates the user's answers, and gives hints if the user requests it. In addition to users, we also have tutors who provide questions and hints. And also examiners who must certify questions to make sure they are not too trivial, and that they are sensible.

Make a use case diagram to model this system. Work out some of your use cases with suitable scenario description.

Design Use Case and Interaction Model

Suppose we want to develop software for an alarm clock. The clock shows the time of day. Using buttons, the user can set the hours and minutes fields individually, and choose between 12 and 24-hour display. It is possible to set one or two alarms. When an alarm fires, it will sound some noise. The user can turn it off, or choose to 'snooze'. If the user does not respond at all, the alarm will turn off itself after 2 minutes. 'Snoozing' means to turn off the sound, but the alarm will fire again after some minutes of delay. This 'snoozing time' is pre-adjustable.

Practice Exercise

Identify the object and describe its attribute values.

ProjectAndTeamStatus : Report

ID = 1

Name = "ProjectReport"

DateOfCreation = "April2015"

SectionName[1] = "Proj Info"

SectionName[2] = "Status Info"

SectionName[3] = "Team Info"

ProjName[1] = "Akash"

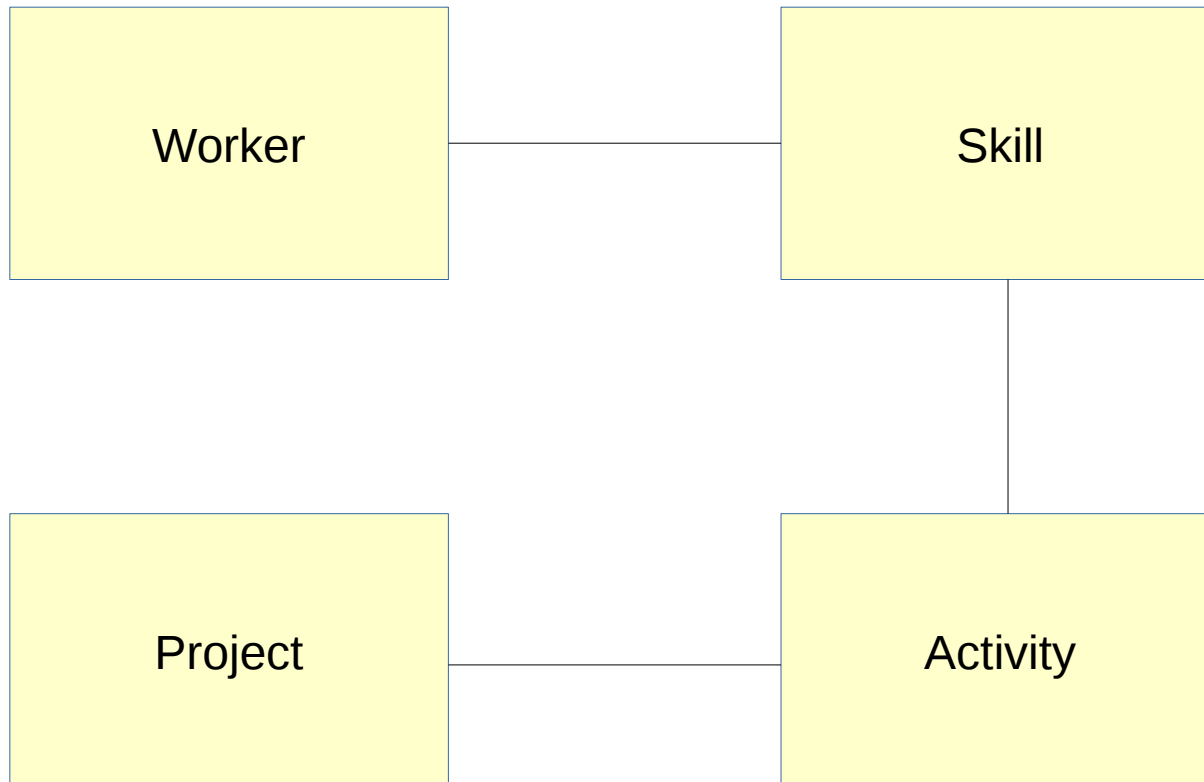
ProjName[2] = "Pawan"

ProjName[3] = "Energy"

OwnerName = "India"

Practice Exercise

Identify the classes and their associations.



Exercise ...cont'd

Update the diagram stepwise to show the following details.

When a worker has a skill, the years of experience is maintained for the relationship.

A worker may have another worker as a manager, and a worker who is a manager must manage five or more workers. Given a manager, you can determine whom she manages, but given a worker, you are unable to determine who his manager is.

An activity may have no more than one previous activity and any number of next activities. Using these rolenames, we can show how activities are ordered. Given an activity, you can only determine its next activities (if it has any) but not its previous activity (if it has one). This is similar to how a team may be made of subteams in that you have to have a team before you can have a subteam.

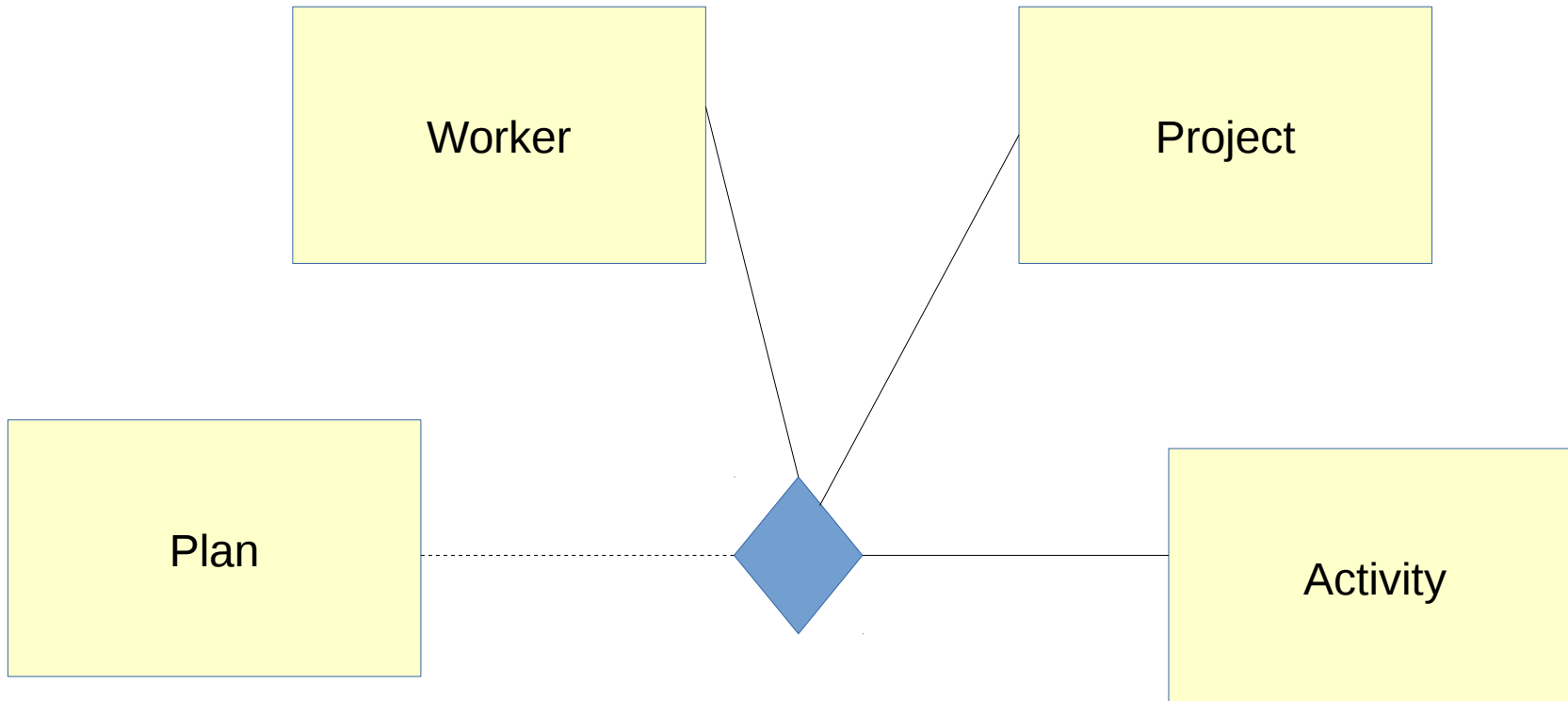
A worker is not simply associated with a set of skills, but a worker has skills. Specifically, a worker must have three or more skills, and any number of workers may have the same skill.

A project is not simply associated with a set of activities, but a project contains activities. Specifically, a project must have one or more activities, and an activity must belong only to one project.

Projects and activities are specific types of work.

Practice Exercise

Identify the classes and their associations.



Exercise ...cont'd

Update the diagram stepwise

A plan pertains to a single project and involves one or more workers and one or more activities.

A worker may have zero or more plans, and a plan must belong to a single worker.

Each plan contains a single schedule. Given a plan, you can determine its schedule, but given a schedule, you are unable to determine the plan to which it pertains.

When an activity is on a schedule, it is called a scheduled activity, and the start date, end date, and duration are maintained for the scheduled activity. Each schedule may have zero or more activities, but an activity must be associated with a single schedule.

A milestone, a major point in project, is a special type of scheduled activity where it has zero or more work products that have a specific status. A milestone has any number of work products associated with it, and a work product may be related to any number of milestones.

Exercise

Restaurant Automation System (RAS):

A restaurant owner wants to computerize his order processing, billing, and accounting activities. He also expects the computer to generate statistical report about sales of different items. A major goal of this computerization is to make supply ordering more accurate so that the problem of excess inventory is avoided as well as the problem of non-availability of ingredients required to satisfy orders for some popular items is minimized. The computer should maintain the prices of all the items and also support changing the prices by the manager. Whenever any item is sold, the sales clerk would enter the item code and the quantity sold. The computer should generate bills whenever food items are sold. Whenever ingredients are issued for preparation of food items, the data is to be entered into the computer. Purchase orders are generated on a daily basis, whenever the stock for any ingredient falls below a threshold value. The computer should calculate the threshold value for each item based on the average consumption of this ingredient for the past three days and assuming that a minimum of two days stock must be maintained for all ingredients. Whenever the ordered ingredients arrive, the invoice data regarding the quantity and price is entered. If sufficient cash balance is available, the computer should print cheques immediately against invoice. Monthly sales receipt and expenses data should be generated whenever the manager would request to see them. The computer should be able to print a menu card containing the menu items and their prices.

1. Prepare a requirements specification for the RAS system. [10]
2. Perform SA/SD based-on your specification obtained in (1). [15]
3. Prepare a UML design model for the RAS system. [15]

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Exercise

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[Req. Doc. As per the discussed template.](#)
2. Perform SA/SD based-on your specification obtained in (1). [15]
[DFD Level 0, 1, 2, Structure chart in sync with \(1\).](#)
3. Prepare a UML design model for the RAS system. [15]
[Use case diagram, class diagram, sequence diagram.](#)

Exercise

Supermarket automation software (SAS):

The manager of a supermarket wants us to develop an automation software. The supermarket stocks a set of items. Customers pick up their desired items from the different counters in required quantities. The customers present these items to the sales clerk. The sales clerk passes the items over a bar code reader and an automatic weighing scale and the data regarding the item type and the quantity get registered.

- SAS should at the end of a sales transaction print the bill containing the serial number of the sales transaction, the name of the item, code number, quantity, unit price, and item price. The bill should indicate the total amount payable.
 - SAS should maintain the inventory of the various items of the supermarket. The manager upon query should be able to see the inventory details. In order to support inventory management, the inventory of an item should be decreased whenever an item is sold. SAS should also support an option by which an employee can update the inventory when new supply arrives.
 - SAS should support printing the sales statistics for every item the supermarket deals with for any particular day or any particular period. The sales statistics should indicate the quantity of an item sold, the price realized, and the profit.
 - The manager of the supermarket should be able to change the the price at which an item is sold as the prices of the different items vary on a day-to-day basis.
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